Notes from the aboveground: Blockchain governance in practice

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EGR 373: Designing Protocols

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Blockchains as a world computer

Blockchain primer

- Blockchains are permissionless, verifiable, immutable computers.
  - → permissionless: "anyone can read or write data."
  - → verifiable: "the data written is authenticated with cryptography."
  - → immutable: "no one can rewrite history."
- Contrast this to AWS
- These properties *enable*:
  1. digital money
  2. "smart" contracts

Smart contract example: peer-to-peer trading

Blockchain primer

- Say Alice has 1 BTC and wants to trade it for 32 ETH.
- She can create a contract `BTC-for-ETH` that does the following:
  1. Escrows her 1 BTC "in" the contract.
  2. If Bob comes along and deposits 32 ETH into the contract, then he can withdraw the 1 BTC.
  3. Then Alice can remove the 32 ETH.
- Trading in the "real-world" is way messier...

Two distinct layers

- Governance *enabled* by blockchains (mostly automated)
  - → "*Blockchains provide a decentralized computational platform. How does this new paradigm allow for automated / decentralized decision-making?*"
  - → **Example:** The DAO as an experiment on decentralized capital formation and collective investment.
- Governance *of* blockchain protocols (mostly human)
  - → "*Blockchains are themselves decentralized protocols. How are these protocols developed and maintained?*"
  - → **Example:** Bitcoin and the coordination effort of rolling out post-quantum cryptography.
- Much like internet protocols / browsers versus applications / products.

Governance *enabled* by blockchains

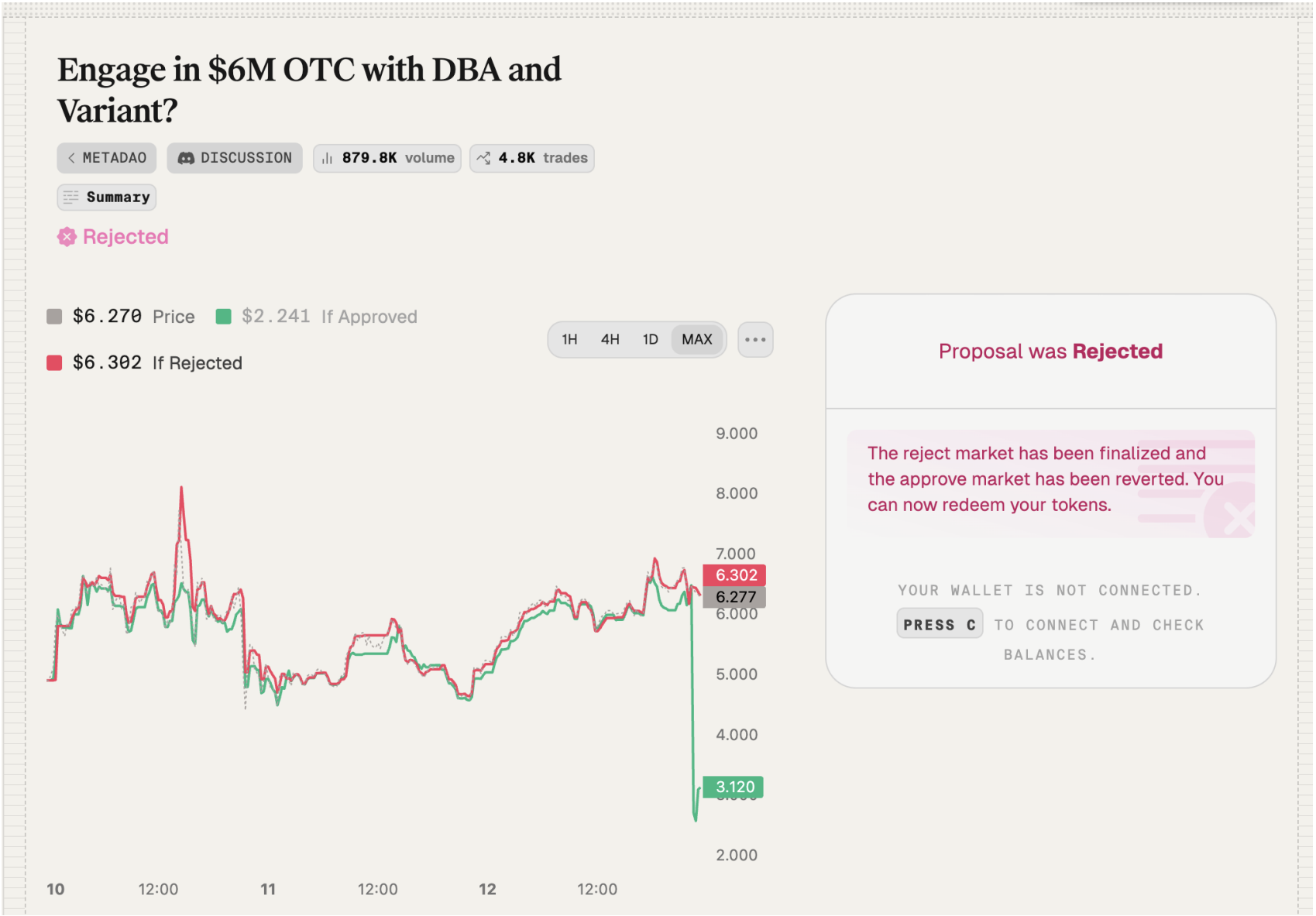
Case study #1: The DAO / MetaDAO

Governance enabled by blockchains

- Goal: Use smart contracts to facilitate capital formation (raising money) and allocation (spending money).
- Key idea: Those who invest money can participate in voting mechanisms to decide how the money is allocated.

Case study #1: The DAO / MetaDAO

Governance enabled by blockchains



Case study #1: The DAO / MetaDAO

Governance enabled by blockchains

- Realities:
  1. Adverse selection
  2. Can encumber startup decision-making speed
- Still very much a greenfield experiment!

Case study #2: Uniswap

Governance enabled by blockchains

- Uniswap: A "decentralized exchange" that runs on a blockchain and allows people to swap tokens (like our example!).
- Goal: A "governance token" (UNI) that represents some notion of ownership over the decentralized protocol.
- Key idea: The token can also accrue value based on the usage of the protocol.

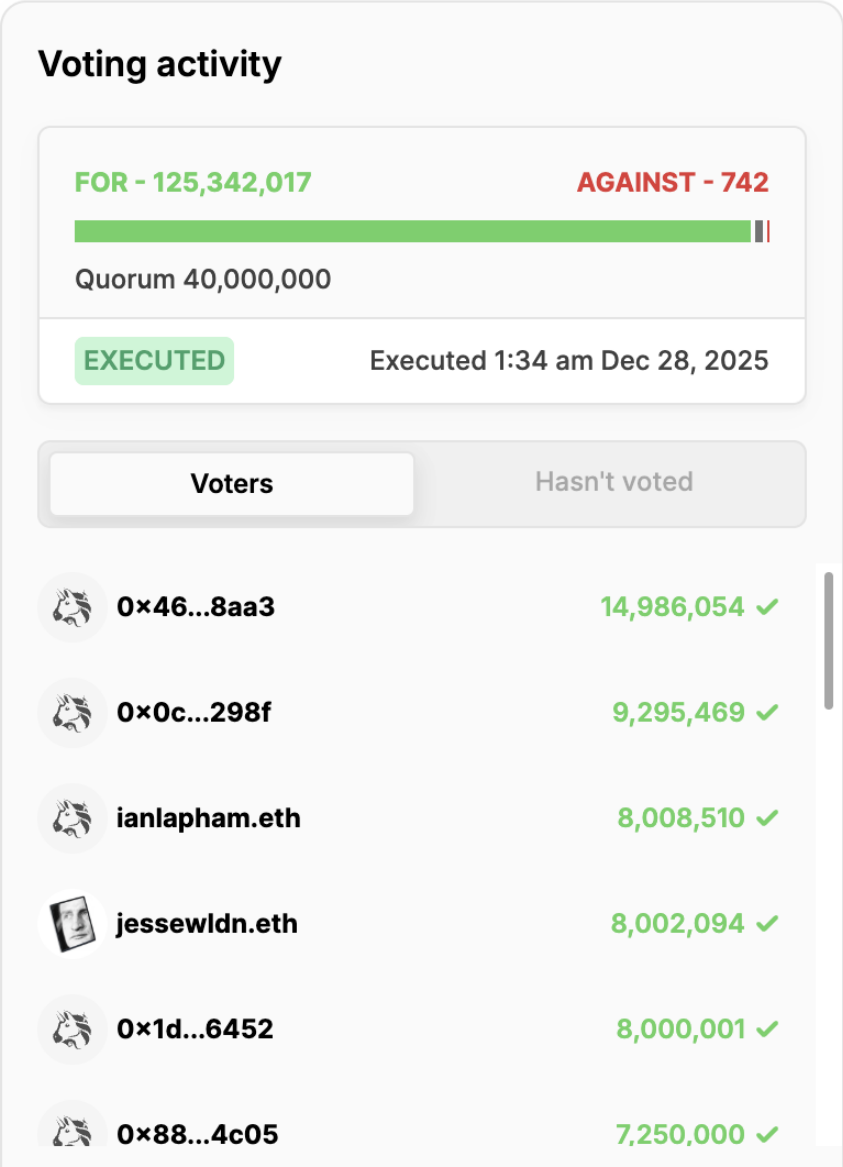
Case study #2: Uniswap

Governance enabled by blockchains

- Realities:
  1. Mismatch between company equity and the UNI token
  2. Voter diffusion and apathy (same with shareholder votes!)
  3. The fee switch saga
- Important note: the regulatory backdrop and concerns about UNI becoming a security.

Case study #2: Uniswap

Governance enabled by blockchains



Case study #3: Constitution DAO

Governance enabled by blockchains

- Goal: Crowd fund to put together a bid on a Sotheby's auction of an original copy of the US Constitution.
- Key idea: Everyone would own a proportional "share" of the document.

Case study #3: Constitution DAO

Governance enabled by blockchains

In this article, we'll cover how ConstitutionDAO used Juicebox to fundraise ~ $46 million (11,500+ ETH) in less than a week.

Nicolas Cage in National Treasure

(Submitted a bid of $43mm USD)

MARKETS

Citadel CEO Ken Griffin pays $43.2 million for Constitution copy, outbidding crypto group

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Case study #3: Constitution DAO

Governance enabled by blockchains

- Realities:
  1. Transparency isn't always good

Governance *of* blockchain protocols

Case study #4: Bitcoin Post-Quantum

Governance of blockchain protocols

- Bitcoin is a decentralized protocol that was launched in 2009.
- Historically, the Bitcoin community has been *extremely* resistant to change anything about the protocol.
- But quantum computers can break the digital signatures underpinning the entire system...
- *What do you do with Satoshi's coins?!* $\leftarrow$ a \$300 billion question.

Case study #5: Ethereum Upgrades and the AllCoreDevs

Governance of blockchain protocols

- Ethereum is immensely different than Bitcoin. Upgrades are (and always were) part of the plan!

Every Ethereum Upgrade

Names are chosen by core developers on AllCoreDevs calls. Execution layer upgrades are named after Devcon host cities, consensus layer upgrades after stars. Combined upgrades merge both into a portmanteau (e.g. Pectra = Prague + Electra).

UPGRADE	EXECUTION LAYER	CONSENSUS LAYER	DATE
Hegota	Bogota Devcon 6	Heze Star	TBD
Glamsterdam	Amsterdam Devconnect	Gloas Star	TBD
Fusaka	Osaka Devcon 5	Fulu Star	Dec 2025
Pectra	Prague Devcon 4	Electra Star	May 2025
Dencun	Cancun Devcon 3	Deneb Star	Mar 2024
Shapella	Shanghai Devcon 2	Capella Star	Apr 2023
The Merge	Paris EthCC	Bellatrix Star	Sep 2022
London	London Devcon 1	— Pre-Beacon	Aug 2021
Berlin	Berlin Devcon 0	—	Apr 2021
Istanbul	Istanbul Historical	—	Dec 2019
Constantinople	— Historical	—	Feb 2019
Byzantium	— Historical	—	Oct 2017
Homestead	—	—	Mar 2016
Frontier	—	—	Jul 2015

Case study #5: Ethereum Upgrades and the AllCoreDevs

Governance of blockchain protocols

- Changes to the Ethereum protocol are proposed to a weekly open-source developer call.

Ethereum Improvement Proposals

All Core Networking Interface ERC Meta Informational

Standards Track: Core

EIP-7594: PeerDAS - Peer Data Availability Sampling

Introducing simple DAS utilizing gossip distribution and peer requests

Authors Danny Ryan (@djrtwo), Dankrad Feist (@dankrad), Francesco D'Amato (@fradamt), Hsiao-Wei Wang (@hwwhww), Alex Stokes (@ralexstokes)

Created 2024-01-12

Requires EIP-4844

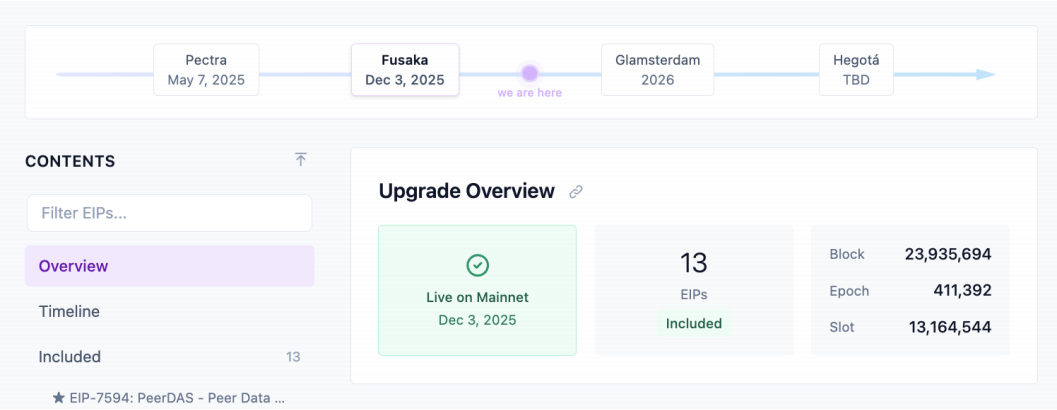
Table of Contents

- Abstract
- Motivation
- Specification
  - Networking
- Rationale
  - Why use DAS to scale the DA layer?
  - What does peer sampling provide?

Case study #5: Ethereum Upgrades and the AllCoreDevs

Governance of blockchain protocols

- The changes are bundled into "hard-forks" at which point all the clients switch to a new version of the software.



- The contents of these hard forks are the subject of intense debate (many stakeholders).

Case study #6: Solana

Governance of blockchain protocols

- Solana is a much newer chain (March 2020) and runs more like a Silicon Valley startup."
- E.g., Toly (the founder) has decision-making power in many cases.
- Still, even Solana is now facing difficulty passing contentious changes to their protocol.

SIMD-0228: Market-Based Emission Mechanism

Proposal failed (61.39% yes of yes+no votes, below 66.67%)

Total Supply: 379050125089013001

Yes Votes: 165236706583125125 (43.5923%) Show Voters

No Votes: 103886399401224841 (27.4070%) Show Voters

Abstain Votes: 12426457071592087 (3.2783%) Show Voters

Not Voted: 97500562033070948 (25.7223%)

In summary

- Blockchains represent a new computational paradigm and have some unique properties.
- Automation and governance can be instantiated *on top* of blockchain protocols.
- The blockchain protocols *themselves* also have quite a rich governance structure.

Thanks!